Al-Ajjaj Limited Company W.L.L

شركة آل عجاج المحدودة ذ.م.م

# COMPANY PROFILE

www.aalcoqatar.com

**A Member Company of RAK Holding**

# Corporate Overview

## About Us

Established in 2005, **Al-Ajjaj Limited Company W.L.L. (AALCO)** is a premier Grade "A" Kahramaa-approved contractor and an ISO 9001, 14001 & 45001 certified engineering powerhouse in Qatar. Backed by elite technical teams, in-house manufacturing, and specialized trading divisions, AALCO delivers self-sufficient turnkey solutions across Extra High Voltage (EHV) power transmission, primary substations, civil infrastructure, potable water pipe networks, strategic dry and wet utilities, applied HVAC/MEP, and industrial manufacturing.

Over two decades of operational rigor, we have built an unshakeable reputation for engineering precision and safety excellence. We are proud to serve Qatar's most prestigious government, healthcare, aviation, energy, and national defense institutions—including Kahramaa, Ashghal, QatarEnergy, Qatari Diar, Hamad International Airport (HIA), Qatar Foundation, Lusail, Supreme Committee, Lekhwiya, Aspire Zone, Qatar Armed Forces, Hamad Medical Corporation (HMC), Qatar Fuel (Woqod), MOEHE, and Al Jazeera Media Network.

**2005**
ESTABLISHED HERITAGE

**Grade "A"**
KAHRAMAA CERTIFIED

## What We Do, Core Sectors & Capabilities

**EHV & HV Transmission:**
33kV, 66kV, 132kV up to 400kV underground cabling, Overhead Lines (OHL), route excavation, jointing, termination, and Kahramaa Hi-Pot testing.

**Primary Substations & Transformers:**
Turnkey construction of indoor/outdoor substations, power transformers up to 500 MVA, 11kV/415V package substations, and control room integration.

**Civil Engineering & Building Works:**
Turnkey construction of substation shells, commercial complexes, industrial warehouses, and Woqod fuel stations, engineered with advanced 3D BIM technology.

**Trenchless HDD & Road Reinstatement:**
Horizontal Directional Drilling (HDD) up to 48", precision road asphalt paving, interlock pavers, and corridor reinstatement compliant with Ashghal.

**Road Asphalt Reinstatement:**
Bulk potable and TSE water transmission networks from DN20 to DN2000 utilizing Ductile Iron (D.I.), HDPE, and copper pipelines.

**Key Infrastructure & Utility Capabilities:**
Potable Water Networks, TSE Networks, Foul Sewer Networks, Stormwater Drainage Networks, Pump Stations & Pumping Systems, Pneumatic Waste Management Networks, and Gas Networks.

**Electro-Mechanical & Applied HVAC:**
Commercial HVAC plants (chillers, VRF/VRV, AHU/FAHU, ducting), industrial steam boilers, hot water calorifiers, and sanitary plumbing systems.

**Smart Metering & AMI Automation:**
Nationwide supply, installation, and integration of smart electricity and water meters, Advanced Metering Infrastructure (AMI), RTUs, and telemetry.

**In-House Switchgear & Control Panels:**
Factory assembly of 11kV switchgear, RMUs, design-verified Form 4b LV switchboards up to 7,100A, feeder pillars, MCCs, and capacitor banks.

**Industrial GI Fabrication & Equipment:**
Custom G.I. substation doors, acoustic louvers, sand trap louvers, meter cabinets, tubular poles, and specialized industrial equipment supply.

# Executive Leadership

## Chairman & Managing Director's Message



***"Since our founding in 2005, our purpose has been resolute: to engineer resilient infrastructure that fuels the growth, ambition, and prosperity of the State of Qatar."***

Since establishing AALCO in 2005, our journey has been guided by technical rigor, integrity, and national pride. Over two decades, we have proudly contributed to the foundational power and civil networks driving Qatar's rapid development. In steadfast alignment with Qatar National Vision 2030, we continue to uphold rigorous institutional governance, sustainable engineering practices, and enduring partnerships that build a lasting legacy of reliability for future generations.

**Engr. Rubaih Mohd A R Al Kubaisi**

## Chief Executive Officer's Message

***"True operational excellence is built on total self-reliance—controlling the entire delivery chain from in-house manufacturing to turnkey field execution."***

At AALCO, operational precision governs every project we execute. From Extra High Voltage transmission and primary substations to civil infrastructure and in-house manufacturing, our strength lies in self-sufficient turnkey delivery. Backed by a strict zero-harm safety culture and advanced engineering methodologies, our teams deliver nation-scale infrastructure on schedule cementing AALCO as Qatar's most dependable EPC partner.

**Engr. Faisal Kunnath**



# EHV & HV Power Transmission

## 33kV • 66kV • 132kV • 220kV • 400kV Turnkey Grid Infrastructure

As an approved Grade "A" Kahramaa Contractor, AALCO delivers nation-scale power transmission networks across Qatar on a complete turnkey EPC basis. Operating across all major voltage thresholds from 33kV and 66kV to 132kV, 220kV, and 400kV, our specialized power division executes complex underground transmission corridors and grid interconnections with uncompromising engineering rigor.

**POWERING A STRONGER QATAR**

## SCOPE OF CAPABILITIES

### Underground EHV & HV Transmission Cabling

Turnkey route excavation, trenching, high-capacity cable pulling, thermal backfilling, protective concrete slabs, and optical fiber pilot integration up to 400kV.

### Specialized Jointing & Terminations

Kahramaa-certified jointers delivering precision straight joints, transition joints, and Gas Insulated Switchgear (GIS) / outdoor transformer terminations for XLPE cables.

### Route Civil & Duct Banks

Deep route excavation, multi-way reinforced concrete duct banks, protective thermal backfill, road crossing sleeves, and pilot fiber duct installation.

### High-Voltage Testing & Pre-Commissioning

Advanced on-site AC resonance testing (Hi-Pot), partial discharge (PD) diagnostics, sheath integrity verification, REF relay calibration, and cable fault location.





# Substations & Power Distribution

## 11kV • 33kV • 66kV • 132kV Turnkey Grid Solutions

**ENERGIZING NATIONAL INFRA**

AALCO delivers complete turnkey EPC solutions for Kahramaa-approved primary and distribution substations across Qatar. From heavy reinforced civil foundations and blast walls to equipment rigging, power transformer installations up to 500 MVA, GIS/AIS switchgear integration, and 33kV/11kV/0.415kV distribution networks, our engineering teams execute critical grid infrastructure with zero-defect commissioning and total national reliability.

## SCOPE OF CAPABILITIES

### Substation EPC Construction

Turnkey construction of indoor Gas Insulated Switchgear (GIS) and outdoor Air Insulated Switchgear (AIS) substations, transformer blast walls, cable cellars, and control buildings.

### Power & Dist. Transformers

Rigging, placement, oil filtration, and commissioning of primary power transformers up to 500 MVA (132kV), and 11/0.433kV oil and dry-type distribution units (500kVA to 1600kVA).

### Overhead Distribution Networks

Construction and maintenance of 33kV, 11kV & 0.415kV overhead lines, pole structures, conductor stringing, pole-mounted transformers, Auto Reclosers, and Air Break Switches (ABS).

### Package Substations & RMU

Installation of 11kV/415V Package Substations (DPS), SF6 Ring Main Units, Drop-Out Fuses (DOF), Fault Passage Indicators (FPI), REF relay testing, and Kahramaa energization.





# Civil Infrastructure & Buildings

## Residential • Commercial • Industrial • Power Projects

AALCO's civil division delivers comprehensive structural concrete engineering, commercial developments, and fast-track infrastructure across Qatar. Utilizing advanced 3D Building Information Modeling (BIM) and extensive in-house plant machinery, our expert teams construct heavy-duty substation shells, logistics warehouses, and specialized fuel retail facilities—complemented by precision full-depth asphalt paving and Ashghal-compliant road corridor reinstatement.

**BUILDING QATAR'S HORIZONS**

## SCOPE OF CAPABILITIES

### Power Projects

Turnkey construction of heavy blast-resistant reinforced concrete walls, transformer foundation beds, basement cable cellars, and multi-story control buildings.

### Residential, Commercial & Fit-Out

Construction of multi-story commercial complexes, residential townships, and institutional buildings driven by advanced 3D BIM modeling and high-end interior fit-outs.

### Industrial Buildings

Turnkey delivery of large-span pre-engineered steel buildings (PEB), heavy logistics depots, industrial workshops, and high-load reinforced concrete floor slabs.

### Specialized Fuel Stations

Complete civil execution of modern petrol retail stations (Woqod standards), canopy structural works, underground fuel tank civil chambers, and retail driveways.






# Water & Wet Utilities Infrastructure

## Pipelines • Potable & TSE Networks • Pumping Stations

**STRATEGIC WATER NETWORKS**

AALCO's infrastructure division delivers comprehensive wet utility engineering and municipal pipeline networks across Qatar. Backed by two decades of proven execution, our specialized teams construct critical utility pipelines ranging from DN20 to DN2000 - encompassing **bulk potable water, TSE transmission, foul sewers, stormwater drainage, and district cooling chilled water networks** compliant with Kahramaa, Ashghal, and QatarEnergy standards.

## SCOPE OF CAPABILITIES

### Potable & TSE Networks

Bulk potable and TSE water transmission networks from DN20 to DN2000, utilizing Ductile Iron (D.I.), HDPE, and reinforced valve chambers.

### Foul Sewer & Stormwater

Turnkey construction of deep gravity foul sewer networks, municipal stormwater drainage, attenuation tanks, outfall structures, and specialized deep precast manhole chambers.

### Chilled Water Networks

Large-diameter pre-insulated carbon steel and HDPE chilled water transmission piping networks, connecting central district cooling plants to commercial, healthcare, and residential developments.

### Pump Stations & Systems

Turnkey civil and electro-mechanical execution of potable water booster pump stations, sewage lifting stations, surge vessels, automated isolation valves, and pneumatic waste networks.





# Electro-Mechanical (MEP) & HVAC

## Central Chillers • VRF Systems • Air Handling • Boilers & Plumbing

AALCO's electro-mechanical division delivers comprehensive commercial HVAC engineering, district cooling integration, and advanced building utilities across Qatar. From central water-cooled chiller plants and specialized ventilation systems to industrial steam boilers and high-capacity sanitary networks, our engineering teams provide complete turnkey design, precision duct fabrication, installation, testing, and commissioning tailored for extreme ambient conditions.

**INTEGRATED CLIMATE ENGINEERING**

## SCOPE OF CAPABILITIES

### Central Chillers & Cooling

Air-cooled and water-cooled chiller plants, district cooling energy transfer stations (ETS), plate heat exchangers, chilled water piping, and primary/secondary pumping systems.

### VRF & Air Handling (AHU)

Multi-split VRV/VRF systems, packaged AC units, Fresh Air Handling Units (FAHU) with energy recovery, industrial exhaust fans, and precision factory-fabricated G.I. ducting.

### Steam Boilers & Calorifiers

Turnkey installation of industrial steam boilers, commercial hot water calorifiers, direct and indirect heating vessels, thermal insulation, and automated pressure controls.

### Plumbing, Drainage & ELV

Potable water supply, stormwater drainage, sewage lifting stations, sanitary fixtures, smart indoor/outdoor lighting controls, small power, and integrated building ELV services.





# Smart Metering & Grid Automation

## AMI • Electricity & Water • SCADA

**SMART GRID AUTOMATION & METERS**

AALCO's smart utility digitalization division delivers nationwide Advanced Metering Infrastructure (AMI) and grid telemetry across Qatar. Executing large-scale deployments for both electricity and water networks, our specialized technical teams provide complete field installation, communication validation, data logging, and SCADA integration—enabling real-time monitoring, automated utility billing, and enhanced operational efficiency compliant with strict Kahramaa standards.

## SCOPE OF CAPABILITIES

### Smart Electricity Meters

Nationwide field installation of high-precision single-phase and three-phase digital smart electricity meters with remote connect/disconnect, anti-tamper sensing, and automated two-way metering communication.

### Smart Water Metering

Supply and deployment of intelligent ultrasonic and electromagnetic smart water meters, automated leak detection telemetry, pulse emitters, and integration with central Kahramaa billing platforms.

### SCADA & RTU Telemetry

Integration of Remote Terminal Units (RTUs), industrial data loggers, cellular/radio communication gateways, and fiber-optic communication networks for real-time substation and utility monitoring.

### Validation & Operations

End-to-end field communication validation, digital signal testing, meter asset verification, service operations, and long-term preventive maintenance agreements for municipal utility networks.





# Integrated In-House Capabilities

## LS Power Systems L.L.C

**LV Switchgear (Siemens & Sivacon S8) & Type Tested:** Up to 7,000A, Form 4b Type 7, 100kA-1s / 65kA-3s, fixed & withdrawable (IEC 61439-1 & 2).

Specialized LVAC and LVDC auxiliary distribution equipment engineered for Kahramaa primary substations.

415V AC and 110V DC switchgear boards, Form 4B, Withdrawable type , IP 43 to IP 54.

**Motor Control Centers (MCC):** Up to Form 4b Type 7, supporting DOL, Star-Delta, soft starters, VFD drives, and ATS with bypass.

**APFC Capacitor Banks:** Automatic power factor correction up to 600 kVAR with 7% detuned harmonic reactors.

**Factory CNC & FAT Test Lab:** CNC sheet punching, copper busbar bending, robotic gasket machine, and 5,000A temperature rise FAT test lab.




## Al Dana Switchgear

**Type-Tested LV Switchgear (Eaton & Lauritz Knudsen / L&T):** Fully type-tested low-voltage main switchboards up to 7,100A, Form 4b, IP43 to IP55.

**Kahramaa LV Feeder Pillars:** 4-Way & 6-Way (1600A & 2500A, DIN & J-Type, 40kA-1sec, IP41/IP54).

**Distribution Package Substations (DPS):** 11kV/415V package substations (500 kVA to 1600 kVA) & SF6 Ring Main Units (RMU).

**Distribution Boards (SMDB & DB):** Sub-Main Boards up to 800A (Form 2b/3b) and Final DBs up to 250A.

**Street Light Feeder Pillars:** 4 to 16 ways up to 250A incoming, 160A outgoing (DEKRA certified 50kA).

**Custom Control Assemblies:** Synchronization panels, HVAC damper control panels, and PLC automation.

**Battery Chargers & LV Current Transformers:** Industrial battery chargers, DC tripping units, and LV current transformers (CTs) for substation protection.



# Integrated In-House Capabilities

## McKinley Factory For Cooling Networks

**Kahramaa Substation Doors & Louvers:** Solid G.I. substation doors, sand-trap louvers, acoustic louvers, roof ventilators, and steel cage ladders.

**Fire-Rated (UL/BS) & Acoustic Doors:** Certified FD120 to FD180 fire-rated steel doors and ASTM E90 acoustic doors (up to 48dB).

**SMACNA Ducting & Ventilation:** Factory CNC-fabricated rectangular/spiral ductwork (DW-144/SMACNA), VCD dampers, and car park jet fans.




## Dana Tradezone W.L.L

**Transformers, Generators & Switchgear:** 132kV power transformers (up to 500 MVA), distribution units, diesel generators, and indoor/outdoor switchgear.

**Cable Accessories & Power Backup:** Cable termination kits, straight/transition joints, industrial UPS (up to 900 kVA), CBS, and 24V–220V DC chargers.

**Piping, Valves & Structural Poles:** **K**ahramaa-approved LDPE copper pipes (BS EN 1057), D.I. valves, tubular streetlight poles, and navigation buoys.




## Nextrabit Technology Solutions

**Enterprise IT & Cloud Solutions:** Scalable Dell enterprise servers, SAN/NAS storage, data center networking, and unified hybrid cloud platforms.

**Cybersecurity & Network Defense:** End-to-end network security, next-gen enterprise firewalls, endpoint protection, and critical data defense.

**Integrated ELV, CCTV & Fiber Cabling:** Turnkey IP surveillance, access control, BMS, and certified multi-core fiber optic structured cabling networks.



# Al-Ajjaj Limited Company W.L.L
# شركة آل عجاج المحدودة ذ.م.م

*"Empowering Infrastructure.*
*Delivering Excellence."*

SCAN ME!


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**Grade "A" Kahramaa Certified • ISO 9001:2015 • ISO 14001:2015 • ISO 45001:2018**